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Agent metadata in metric views

Agent metadata (also known as semantic metadata) enhances data visualization and improves large language model (LLM) accuracy by providing display names, format specifications, and synonyms that give business context to your metrics. This metadata helps visualization tools and natural language tools like [Genie Spaces](#) interpret and work with your data more effectively.

NOTE

Requires Databricks Runtime 17.3 and YAML version 1.1. See [version requirements](#).

What is agent metadata?

Agent metadata includes display names, format specifications, and synonyms that provide additional context. This metadata helps visualization tools, such as AI/BI dashboards, and natural language tools, such as [Genie Spaces](#), interpret and work with your data more effectively. Agent metadata is defined in the metric view YAML definition.

NOTE

When you create or alter metric views with specification version 1.1, any single-line comments (denoted with #) in the YAML definition are removed when the definition is saved. See [Upgrade to YAML 1.1](#) for options and recommendations when upgrading existing YAML definitions.

The examples on this page use the TPC-H sample dataset (`samples.tpch.orders`), which is available by default in [Unity Catalog datasets](#). The TPC-H dataset is a wholesale supply chain with tables for orders, customers, suppliers, and

 Ask Genie

names in the `orders` table use the `o_` prefix (for example, `o_orderdate` for order date, `o_totalprice` for total price). For details about the TPC-H schema and data model, see [Tutorial: Build a complete metric view with joins](#).

Display names

Display names provide human-readable labels that appear in visualization tools instead of technical column names. Display names are limited to 255 characters.

The following example shows display names defined on the `order_date` dimension (tracking when orders were placed) and `total_revenue` measure (calculating the sum of all order prices).

YAML

```
version: 1.1
source: samples.tpch.orders

dimensions:
  - name: order_date
    expr: o_orderdate
    display_name: 'Order Date'

measures:
  - name: total_revenue
    expr: SUM(o_totalprice)
    display_name: 'Total Revenue'
```

Synonyms

Synonyms help LLM tools, such as [Genie](#), discover dimensions and measures through user input by providing alternative names. You can define synonyms using either block style or flow style YAML. Each dimension or measure can have up to 10 synonyms. Each synonym is limited to 255 characters.

The following example shows synonyms defined on the `order_date` dimension (when orders were placed) and `total_revenue` measure (sum of all order prices). The

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synonyms allow users to ask questions using natural language such as "show me revenue by order time" or "what are total sales by date of order":

YAML

```
version: 1.1
source: samples.tpch.orders

dimensions:
  - name: order_date
    expr: o_orderdate
    # block style
    synonyms:
      - 'order time'
      - 'date of order'

measures:
  - name: total_revenue
    expr: SUM(o_totalprice)
    # flow style
    synonyms: ['revenue', 'total sales']
```

Format specifications

Format specifications define how values should be displayed in visualization tools. The following tables include supported format types and examples.

Numeric formats

Format Type	Required Options	Optional Options
Number: Use plain number format for general numeric values with optional decimal place control and abbreviation options.	<code>type:</code> <code>number</code>	<code>decimal_places</code>: Controls the number of places shown after the decimal. <code>type</code>: (Required if <code>decimal_places</code> is specified) <code>max</code> <code>exact</code>  Ask Genie

Format Type	Required Options	Optional Options
		▪ <code>all</code> ◦ <code>places</code>: Integer value from 0–10 (required if type is <code>max</code> or <code>exact</code>) • <code>hide_group_separator</code>: When set to true, removes any applicable number grouping separator, such as a <code>,</code>. ◦ <code>true</code> ◦ <code>false</code> • <code>abbreviation</code>: ◦ <code>none</code> ◦ <code>compact</code> ◦ <code>scientific</code>
Currency: Use currency format for monetary values with ISO-4217 currency codes.	<code>type:</code> <code>currency</code>	• <code>currency_code</code>: ISO-4217 code (required). For example, the following codes insert the symbol for US dollars, Euros, and Yen, respectively. ◦ <code>USD</code> ◦ <code>EUR</code> ◦ <code>JPY</code> • <code>decimal_places</code>: Controls the number of places shown after the decimal. ◦ <code>type</code>: (Required if <code>decimal_places</code> is specified) ▪ <code>max</code> ▪ <code>exact</code> ▪ <code>all</code> • <code>hide_group_separator</code>: When set to true, removes any applicable

Format Type	Required Options	Optional Options
		number grouping separator. ◦ <code>true</code> ◦ <code>false</code> • <code>abbreviation</code>: ◦ <code>none</code> ◦ <code>compact</code> ◦ <code>scientific</code>
Percentage: Use percentage format for ratio values expressed as percentages.	<code>type:</code> <code>percentage</code>	• <code>decimal_places</code>: Controls the number of places shown after the decimal. ◦ <code>type</code>: (Required if <code>decimal_places</code> is specified) ▪ <code>max</code> ▪ <code>exact</code> ▪ <code>all</code> • <code>hide_group_separator</code>: When set to true, removes any applicable number grouping separator. ◦ <code>true</code> ◦ <code>false</code>
Byte: Use byte format for data size values displayed with appropriate byte units (KB, MB, GB, etc.).	<code>type:</code> <code>byte</code>	• <code>decimal_places</code>: Controls the number of places shown after the decimal. ◦ <code>type</code>: (Required if <code>decimal_places</code> is specified) ▪ <code>max</code> ▪ <code>exact</code> ▪ <code>all</code> ◦ <code>places</code>: Integer value from 0–10 (required if type is <code>max</code> or <code>exact</code>)

Format Type	Required Options	Optional Options
		<code>hide_group_separator</code>: When set to true, removes any applicable number grouping separator. <code>true</code> <code>false</code>

Numeric formatting examples

Number

Currency

Percentage

Byte

YAML

```
format:
  type: number
  decimal_places:
    type: max
    places: 2
  hide_group_separator: false
  abbreviation: compact
```

Date and time formats

The following table explains how to work with date and time formats.

Format Type	Required Options	Optional Options
Date: Use date format for date values with various display options.	<code>type: date</code> <code>date_format</code>: Controls the way the date is displayed <code>locale_short_month</code>: Displays the date with an abbreviated month <code>locale_long_month</code>: Displays the date with the full name of the month <code>year_month_day</code>: Formats the date as YYYY-MM-DD <code>locale_number_month</code>: Displays the date with a month as a number <code>year_week</code>: Formats the date as a year and a week number. For example, <code>2025-W1</code>	<code>leading_zeros</code>: Controls whether single digit numbers are preceded by a zero <code>true</code> <code>false</code>
DateTime: Use datetime format for timestamp values combining date and time.	<code>type: date_time</code> <code>date_format</code>: Controls the way the date is displayed <code>no_date</code>: Date is hidden <code>locale_short_month</code>: Displays the date with an abbreviated month <code>locale_long_month</code>: Displays the date with the full name of the month <code>year_month_day</code>: Formats the date as YYYY-MM-DD <code>locale_number_month</code>: Displays the date with a month as a number	<code>leading_zeros</code>: Controls whether single digit numbers are preceded by a zero <code>true</code> <code>false</code>

Format Type	Required Options	Optional Options
	◦ <code>year_week</code>: Formats the date as a year and a week number. For example, <code>2025-W1</code> • <code>time_format</code>: ◦ <code>no_time</code>: Time is hidden ◦ <code>locale_hour_minute</code>: Displays the hour and minute ◦ <code>locale_hour_minute_second</code>: Displays the hour, minute, and second	

NOTE

When working with a `date_time` type, at least one of `date_format` or `time_format` must specify a value other than `no_date` or `no_time`.

Datetime formatting examples

Date **DateTime**

YAML

```
format:
  type: date
  date_format: year_month_day
  leading_zeros: true
```

Downstream tools integration

Semantic metadata automatically populates downstream tools that consume the metric view:

- **AI/BI dashboards:** Display names and format specifications are automatically populated in dashboard datasets and visualizations to improve dashboard readability.
- **Genie Spaces:** Synonyms are automatically imported to help Genie better discover and understand available dimensions and measures from the metric view.

Complete example

The following example shows a metric view definition that tracks sales performance and includes all agent metadata types. The metric view analyzes order data to calculate revenue metrics, segment customers by order value, and track order volumes.

Customer segments are defined as follows:

- Enterprise: Orders over \$100,000
- Mid-market: Orders between \$10,000 and \$100,000
- Small and medium-sized businesses: Orders under \$10,000

The metadata supports natural language queries such as "show me total sales by customer segment" or "what's the average revenue per order."

YAML

```
version: 1.1
source: samples.tpch.orders
comment: Comprehensive sales metrics with enhanced semantic metadata
dimensions:
  - name: order_date
    expr: o_orderdate
    comment: Date when the order was placed
    display_name: Order Date
    format:
      type: date
      date_format: year_month_day
      leading_zeros: true
    synonyms:
      - order time
      - date of order
```

```

- name: customer_segment
  expr: |
    CASE
      WHEN o_totalprice > 100000 THEN 'Enterprise'
      WHEN o_totalprice > 10000 THEN 'Mid-market'
      ELSE 'SMB'
    END
  comment: Customer classification based on order value
  display_name: Customer Segment
  synonyms:
    - segment
    - customer tier
measures:
- name: total_revenue
  expr: SUM(o_totalprice)
  comment: Total revenue from all orders
  display_name: Total Revenue
  format:
    type: currency
    currency_code: USD
    decimal_places:
      type: exact
      places: 2
    hide_group_separator: false
    abbreviation: compact
  synonyms:
    - revenue
    - total sales
    - sales amount
- name: order_count
  expr: COUNT(1)
  comment: Total number of orders
  display_name: Order Count
  format:
    type: number
    decimal_places:
      type: all
    hide_group_separator: true
  synonyms:
    - count
    - number of orders
- name: avg_order_value
  expr: SUM(o_totalprice) / COUNT(1)
  comment: Average revenue per order
  display_name: Average Order Value
  format:
    type: currency
    currency_code: USD
    decimal_places:
      type: exact

```

places: 2
synonyms:
- aov
- average revenue

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